Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS DELIVERY

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Summary

Healthcare Data Consultant with 10+ years of experience leading the technical delivery and governance of analytics, data platforms, and ML solutions across clinical, operational, and revenue-cycle domains. Combines hands-on expertise in Python, SQL, Snowflake, SQL Server, AWS, Azure, and production data pipelines with leadership in data quality, source-to-target reconciliation, documentation, and shared standards. Skilled at turning ambiguous healthcare challenges into trusted analysis, reusable data models, real-time services, and stakeholder-ready reporting supported by testing, CI/CD, and observability. Known for aligning engineering, data science, operations, and business stakeholders while establishing transparent, scalable, and well-governed solutions for regulated environments.

Experience

HHAeXchange
Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox

HEALTHCARE DATA INTEGRATION CONSULTANT — CONTRACT VIA VEERSA TECHNOLOGIES *Oct. 2025 - Jun. 2026*

Led healthcare source analysis, data-quality investigation and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains, reducing client-specific code by 37%.
- Created or materially redesigned 30 SQL/dbt data-quality tests and designed macros to automate null, duplicate, referential-integrity, business-rule, lifecycle, and source-to-target checks.
- Profiled source schemas and converted ambiguous business rules into documented mappings, acceptance criteria, and onboarding standards, plus 12 data-quality runbooks that standardized investigation and onboarding practices.

Adonis
AWS: S3, Snowflake, SQLMesh, SQL, Python

HEALTHCARE REVENUE CYCLE DATA CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS *Mar. 2025 - Jul. 2025*

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models spanning four source entities for Hopebridge, supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated validation across 54 SQLMesh models, including exact row-count audits, uniqueness checks, and not-null; achieved a 100% pass rate across 117+ configured audit applications at final delivery, including exact row-count reconciliation for 40 models, with no unresolved discrepancies.
- Operationalized 54 SQLMesh models with schema-change detection, audit logging, CI release gates, and reproducible recovery procedures, enabling the FTE team to independently deploy and maintain the pipeline after handoff.
- Authored implementation playbooks, plus internal handoff documentation and a QA runbook that established a repeatable healthcare-data validation workflow for an engineering team new to the domain.

GrowData Analytics *Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec*
INDEPENDENT DATA AUTOMATION & ANALYTICS CONSULTANT *Sep. 2024 - Oct. 2025*
Built a document-to-data product that converted semi-structured invoices into validated, auditable transaction records.

- Engineered a Ruby, Tesseract OCR, and LLM pipeline that converted handwritten and printed invoices into structured transactions, using cross-field validation, reconciliation, and confidence routing to achieve [X%] extraction accuracy across [N invoices].
- Added RSpec-backed validation, structured audit logs, error taxonomies, retry controls, and usage reporting, reducing unrecoverable processing failures by [X%] and mean time to resolution by [Y%].

Arkos Health *AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter*
SENIOR DATA ANALYST *Jun. 2023 - Jan. 2025*
Delivered healthcare analytics, recurring reporting and data-quality investigations while improving the governed data foundation used by the BI team.

- Used SQL across Athena and Redshift to reconcile membership and demographic reporting, isolating differences in extraction criteria, eligibility rules, distinct-member logic, and duplicate handling across [N members] and reducing reporting discrepancies by [X%].
- Established a governed repository for scheduled reports, ad hoc SQL, dbt tests, and Jupyter notebooks; introduced peer-review and documentation standards that reduced duplicated logic by [X%] and analyst onboarding time by [Y%].

Merative *Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps*
SENIOR DATA SCIENTIST *Jul. 2022 - Mar. 2023*
Led data validation, compliance and migration assurance for a regulated healthcare platform modernization.

- Developed automated Python and SQL regression tests for Azure Data Factory and Synapse pipelines, detecting schema drift, transformation defects, and record-level fidelity issues before release and reducing migration defects by [X%].
- Designed and executed HIPAA-compliant de-identification validation across more than 200 million patient records in Azure, verifying [N PHI fields and rules] and resolving [X%] of exceptions before HITRUST readiness review.

IBM *Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle*

SENIOR DATA SCIENTIST *Nov. 2021 - Jul. 2022*
Delivered client analytics and large-scale migration validation for healthcare administrative data.

- Built Spark, Hive, and Impala analytics for administrative healthcare data, translating [N business requirements] into validated reports and analytical outputs while reducing delivery time by [X%].
- Developed distributed migration-validation analyses across more than 150 billion patient records, identifying [N data-fidelity defects] before production transition and preventing [X downstream incidents].

SENIOR DATA ARCHITECT *Dec. 2020 - Nov. 2021*
Led clinical-data integration and unstructured-text analysis at enterprise scale.

- Architected Spark and Hadoop pipelines that integrated and normalized more than 100 billion structured and semi-structured clinical records for [N downstream analytics and application workflows], meeting [X%] of processing SLAs.
- Built Python and Spark NLP workflows to analyze nearly six million provider notes, extracting [N reusable themes or features] and reducing manual text-review effort by [X hours or X%].

DATA ARCHITECT *Nov. 2017 - Dec. 2020*
Owned client-facing integration, discrepancy resolution and data-quality improvement across multi-organization healthcare networks.

- Designed standardized SQL validation frameworks and client-review gates that reduced validation rework by 25% and mapping rework by 45% across [N implementations].
- Traced data and application discrepancies across a 73-organization healthcare network from source ingestion through production, documenting defects in Jira and reducing mean time to resolution by [X%].

Software Developer, Data Platform *Sep. 2016 - Nov. 2017*

Built internal data-curation and knowledge-transfer tools that reduced costs and onboarding effort.

- Built an internal data-curation application that replaced a $30,000 annual vendor contract, then rewrote technical runbooks to reduce knowledge-transfer time by 80% for [N engineers or client teams].

Data Scientist *Nov. 2015 - Sep. 2016*

Coordinated client integration delivery, contractor work and process automation.

- Directed technical delivery across 36 client data-integration projects and established code-review, QA, and issue-resolution standards for 12 remote contractors, improving on-time acceptance by [X%] and reducing escaped defects by [Y%].
- Developed Python, Ruby, and SQL automation that eliminated 28% of unnecessary development work; authored more than 125 technical knowledge-base articles, approximately 30% of the team's online content, to standardize implementation practices.

Explorys *Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins*

Data Science Consultant — Contract via TEKsystems *Jun. 2015 - Nov. 2015*

Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Diagnosed production Hadoop and Oracle ingestion failures and implemented Jenkins-backed scheduling controls across more than 10 client data warehouses, increasing successful job completion to [X%] and reducing recurring incidents by [Y%].

AmTrust Financial Services *VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL*

Software Engineer *Mar. 2014 - Jun. 2015*

Developed MVC components and web forms for insurance policy software.

- Designed and implemented C and ASP.NET MVC components and web forms for an insurance policy platform and new product offering, delivering [N production features] with [X%] fewer post-release defects.
- Profiled more than 65 rating forms, isolated C and SQL performance bottlenecks, and reduced median processing time from [A seconds] to [B seconds].
- Established automated unit testing for multi-tier C and ASP.NET MVC components, increasing code coverage from [A%] to [B%] and reducing regression defects by [X%].

PPG Industries *VB.NET, C#, SQL, C++, FORTRAN*

Color Scientist *Apr. 2012 - Jun. 2013*

Developed numerical methods and software for automotive color matching and measurement.

- Developed C++ and FORTRAN numerical classification algorithms that improved automotive color-matching accuracy by 40%, validated against [N laboratory samples].
- Engineered VB.NET, C, and SQL software to correlate measurements across [N instruments], reducing inter-instrument variance by [X%] and improving QA repeatability by [Y%].
- Developed numerical formulation and pigment-analysis methods for an automated automotive color-matching facility, processing [N formulations per day] and reducing manual analysis by [X hours per week].

Multiple Institutions

Adjunct Professor *Aug. 2010 - May. 2019*

Taught physics & astronomy lecture series and lab courses.

- Designed and delivered quantitative physics and astronomy curricula for [N courses and N students], using interactive, peer-led, and just-in-time methods to improve assessment scores by [X%].
- Produced more than 700 hours of recorded lectures and laboratory material using [LMS and recording tools], enabling on-demand instruction for [N students] with a [X%] completion rate.
- Analyzed evaluation and learning-outcome data to refine course delivery, exceeding institutional benchmarks by 16%-17% year over year.

Professional Recognition

HDM, SPECIAL RECOGNITION (*2025*), Recognized for exceptional performance and commitment in client delivery.
Arkos, ARKOS HEALTH BI STAR (*2023*), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.
Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (*2022*), Recognized for systems migration efforts.
IBM, GOLD CHAMPION (*2018 - 2022*), Recognized for 1,477 classroom hours delivered and 43 certificates earned.
IBM, LAB SERVICES AWARD (*2018*), Recognized for coordinating internal teams to deliver solutions on time.
IBM, EMINENCE AND EXCELLENCE AWARD (*2017*), Recognized for developing a new data curation tool.
PPG, COLOR LAB AUTOMATION AWARD (*2012*), Recognized for expanding data collection and analysis reporting.

Volunteer

HSEA
ETL BRIGADE — *Feb. 2025 - May. 2025*

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll
INSTRUCTOR — *Oct. 2018 - May. 2019*

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE
COACH — *Mar. 2017 - Mar. 2019*

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America
ETL GUILD — *Jan. 2015 - Jun. 2015*

The initiative was brought about to apply data science tools to improve and expand public record access.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization — *Dec. 2020*
"An algorithm uses past utilization and current waitlist info to predict surgery duration. — IBM

REACH Initiative: IBM Watson Care Insights — *2019*
Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow. — IBM

Content & Data Analytics Learning Exchange — *2018*
Introduction of implementation of machine learning algorithms using real-world health data analytics. — IBM

2018 Innovation Research Interchange Fall Networks Conference — *Sep. 2018*
Using NLP to extract insight from unstructured data and transform it from a data lake. — IRI

Cross Team Training Series — *2016*
A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes. — EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets — *Aug. 2025*
Reproducible Postgres load of Tuva seed datasets with robust validation. — HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction [a] — *Dec. 2020*
Improvement on traditional fixed ratio methods for total procedure time estimation. — IBM

Process and Requirements for CCD Data Integration — *Oct. 2017*
Processes and requirements for overcoming the many difficulties with CCD data curation. — IBM

AI Paint Formula Prediction from Spectroscopic Measurements [b] *May. 2015*
Naive bayesian color and effects matching algorithm of complex automotive paint formula. SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching *Feb. 2013*
Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching. USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy [c] *Dec. 2010*
Spectroscopic and Photometric analysis of two open clusters in the galactic plane. OhioLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107 *May. 2008*
Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX. VizieR

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107 *Feb. 2008*
New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107). PASP

A Novel Method for Low-Toxic Photo-Etch Printing [d] *May. 2003*
Application of aquatint etching techniques to photo-etch procedures. MAPC

Education

2022	**IBM Data Science Professional Certificate** [a]	*IBM*
2015	**Certificate of Specialization** Data Science[b]	*Johns Hopkins University*
2010	**Master of Science** Astrophysics[c]	*University of Cincinnati*
2004	**Bachelor of Fine Arts** Printmaking[d]	*Ohio University*